Framework of n-Overarching Axioms: Forms and Parameterizations

Independent Researcher

August 14, 2026

1 Introduction

This document outlines a subset of n-overarching axioms, specifically defining the concepts of "n-forms" and "proper n-forms." While the mechanical behaviors of these forms share parallels with established algebraic concepts (such as exact parameterizations, basis vectors, and n-ary quasigroups), the axiomatic structure, categorization, and nomenclature presented here constitute a novel classificatory framework.

2 The n-Form

An n-**form** is defined as a 1-overarching axiom that establishes a strict, unique parameterization of a set using external variables.

Let x be an element of a primary set S. Let the input variables $y_1, y_2, \ldots, y_n$ belong to distinct, external sets $A_1, A_2, \ldots, A_n$. An n-form dictates that there exists a function $f : A_1 \times A_2 \times \cdots \times A_n \to S$ such that:

$$x = f(y_1, y_2, \ldots, y_n)$$

2.1 Properties of the n-Form

For an n-form to be valid, it must satisfy the following uniqueness constraint:

- **Unique Tuple Designation:** For every element $x \in S$, there exists exactly one unique ordered tuple $(y_1, y_2, \ldots, y_n)$ that evaluates to x. No two distinct elements in S share the same tuple of inputs.
- **Variable Sharing:** While the tuple as a whole is strictly unique to x, the individual specific variables y_i within the tuple are not required to be uniquely bound to x.

3 The Proper n-Form

A **proper n-form** is a stricter subset of the n-form that enforces global algebraic isolation and dimensional independence.

An n-form is considered "proper" if and only if it satisfies two additional overarching properties:

3.1 Strict Non-Uniqueness of Variables

For every variable y_i in the function f, y_i is **strictly not unique** to the output x. This ensures that every variable acts as a fully independent dimension or parameter, preventing the geometric space from collapsing.

3.2 Omni-Directional Invertibility (The g_i Functions)

For every input variable y_i, there exists an exact function g_i that perfectly isolates y_i without a loss of uniqueness.

Specifically, for each variable $y_1, y_2, \ldots, y_n$, there exist functions $g_1, g_2, \ldots, g_n$ such that:

$$\begin{aligned}
y_1 &= g_1(x, y_2, \ldots, y_n) \\
y_2 &= g_2(x, y_1, y_3, \ldots, y_n) \\
&\cdots \\
y_n &= g_n(x, y_1, y_2, \ldots, y_{n-1})
\end{aligned}$$

Because the output x dictates the entirety of the original unique tuple, the presence of x in the g_i functions acts as a "master key," meaning each g_i function guarantees a singular, unique output for y_i across all possible valid combinations.